

## Healthcare\_Silver\_layer:

```
1 .....silver_layer.....
2
3 CREATE OR REPLACE TABLE SILVER_PATIENTS AS
4 SELECT
5     TRIM(Patient_ID)::VARCHAR AS Patient_ID,
6     UPPER(TRIM(Gender)) AS Gender,
7     UPPER(TRIM(AGE_GROUP)) AS AGE_GROUP
8 FROM BRONZE.PATIENTS
9 WHERE Patient_ID IS NOT NULL;
10
11 CREATE OR REPLACE TABLE SILVER_ADMISSIONS AS
12 SELECT
13     UPPER(admission_id) AS admission_id,
14     UPPER(patient_id) AS patient_id,
15     admission_date AS admission_date,
16     discharge_date AS discharge_date,
17     DATEDIFF('day', admission_date, discharge_date) AS length_of_stay,
18     UPPER(department) AS department,
19     UPPER(hospital) AS hospital
20 FROM BRONZE.ADMISSIONS
21 WHERE admission_id IS NOT NULL
22 AND patient_id IS NOT NULL
23 AND admission_date IS NOT NULL
24 AND discharge_date IS NOT NULL
25 AND admission_date
26 discharge_date;
27
28 CREATE OR REPLACE TABLE SILVER_DIAGNOSIS AS
29 SELECT
30     TRIM(Admission_ID)::VARCHAR AS Admission_ID,
31     TRIM(Diagnosis_Category) AS Diagnosis_category
32 FROM BRONZE.DIAGNOSIS;
33
34 CREATE OR REPLACE TABLE SILVER BILLING AS
35 SELECT
36     TRIM(Admission_ID)::VARCHAR AS Admission_ID,
37     TRIM(Treatment_cost) AS TREATMENT_COST
38 FROM BRONZE.BILLING;
39
```

## Gold\_Layer:

```
40 .....gold_layer.....
41
42 CREATE OR REPLACE TABLE gold.dim_diagnosis (
43     diagnosis_key INTEGER AUTOINCREMENT,
44     diagnosis_category STRING
45 );
46
47
48 INSERT INTO gold.dim_diagnosis (diagnosis_category)
49 SELECT DISTINCT diagnosis_category
50 FROM new_hackthon.silver.silver_diagnosis;
51
52
53 select * from dim_diagnosis;
54
55 -----dim_department.....
56 CREATE OR REPLACE TABLE gold.dim_department (
57     department_key INTEGER AUTOINCREMENT,
58     department_name STRING
59 );
60
61 INSERT INTO gold.dim_department (department_name)
62 SELECT DISTINCT department
63 FROM new_hackthon.silver.silver_admissions;
64
65
66 select * from dim_department;
67
68 -----dim_hospital.....
69 CREATE OR REPLACE TABLE gold.dim_hospital (
70     hospital_key INTEGER AUTOINCREMENT,
71     hospital_name STRING
72 );
73
74
75 INSERT INTO gold.dim_hospital (hospital_name)
76 SELECT DISTINCT hospital
77 FROM new_hackthon.silver.silver_admissions;
78
```

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80 -----dim_patient.....
81 CREATE OR REPLACE TABLE gold.dim_patients (
82     patient_key    INTEGER AUTOINCREMENT,
83     patient_id     STRING,
84     gender         STRING,
85     age_group      STRING,
86     e_start_date   DATE,
87     e_end_date     DATE,
88     is_active      BOOLEAN
89 );
90
91 UPDATE gold.dim_patients tgt
92 SET
93     e_end_date = CURRENT_DATE - 1,
94     is_active = FALSE
95 FROM new_hackthon.silver.silver_patients src
96 WHERE tgt.patient_id = src.patient_id
97 AND tgt.is_active = TRUE
98 AND (tgt.gender <> src.gender OR tgt.age_group <> src.age_group);
99
100 INSERT INTO gold.dim_patients (
101     patient_id, gender, age_group,
102     e_start_date, e_end_date, is_active
103 )
104 SELECT
105     patient_id,
106     gender,
107     age_group,
108     CURRENT_DATE,
109     '9999-12-31',
110     TRUE
111 FROM new_hackthon.silver.silver_patients;
112
113
114 -----dim_date.....
115 CREATE OR REPLACE TABLE gold.dim_date (
116     date_key INTEGER AUTOINCREMENT,
117     full_date DATE,
118     day_name STRING,
119     month_name STRING,
120     year_number INTEGER
121 );
122
123 INSERT INTO gold.dim_date (full_date, day_name, month_name, year_number)
124 SELECT DISTINCT
125     admission_date,
126     DAYNAME(admission_date),
127     MONTHNAME(admission_date),
128     YEAR(admission_date)
129 FROM new_hackthon.silver.silver_admissions;
130
131 select * from gold.dim_date;

```

Fact\_table:

```

133 -----fact table.....
134 CREATE OR REPLACE TABLE gold.fact_admission (
135     admission_id    STRING,
136
137     patient_key     INTEGER, -- from dim_patients (SCD2)
138     department_key  INTEGER, -- from dim_department
139     hospital_key    INTEGER, -- from dim_hospital
140     diagnosis_key   INTEGER, -- from dim_diagnosis
141     date_key        INTEGER, -- from dim_date (admission date)
142
143     treatment_cost  NUMBER,
144
145     length_of_stay  NUMBER, -- discharge - admission
146     readmission_count NUMBER -- per patient
147 );
148
149
150 INSERT INTO gold.fact_admission
151 SELECT
152     a.admission_id,
153
154     p.patient_key,
155     d.department_key,
156     h.hospital_key,
157     dg.diagnosis_key,
158     dt.date_key,
159
160     b.treatment_cost,
161
162     DATEDIFF(
163         'day',
164         a.admission_date,
165         a.discharge_date
166     ) AS length_of_stay,
167
168     COUNT(a.admission_id)
169     OVER (PARTITION BY a.patient_id) - 1
170     AS readmission_count
171 FROM new_hackthon.silver.silver_admissions a

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```

172 FROM new_hackthon.silver.silver_admissions a
173
174 JOIN new_hackthon.silver.silver_billing b
175     ON a.admission_id = b.admission_id
176
177 JOIN new_hackthon.silver.silver_diagnosis sd
178     ON a.admission_id = sd.admission_id
179
180 -- * DIM JOINS
181 JOIN gold.dim_department d
182     ON a.department = d.department_name
183
184 JOIN gold.dim_hospital h
185     ON a.hospital = h.hospital_name
186
187 JOIN gold.dim_diagnosis dg
188     ON sd.diagnosis_category = dg.diagnosis_category
189
190 JOIN gold.dim_date dt
191     ON a.admission_date = dt.full_date
192
193 -- * SCD-2 patient (ONLY ACTIVE RECORD)
194 JOIN gold.dim_patients p
195     ON a.patient_id = p.patient_id
196     AND p.is_active = TRUE;
197
198 select * from gold.fact_admission;
199
200
201

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## Kpi \_ business\_requiriment:

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201 -----READMISSION COUNT (Patient-wise)-----
202
203 SELECT
204     patient_key,
205     count(readmission_count) AS total_readmissions
206 FROM gold.fact_admission
207 GROUP BY patient_key
208 ORDER BY total_readmissions DESC;
209
210 -----AVERAGE LENGTH OF STAY (Overall)-----
211 SELECT ROUND(AVG(length_of_stay), 2) AS avg_length_of_stay_days
212 FROM gold.fact_admission;
213
214 -----DEPARTMENT-WISE PERFORMANCE-----
215
216 SELECT
217     d.department_name,
218     COUNT(f.admission_id) AS total_admissions,
219     ROUND(AVG(f.length_of_stay), 2) AS avg_length_of_stay
220 FROM gold.fact_admission f
221 JOIN gold.dim_department d
222     ON f.department_key = d.department_key
223 GROUP BY d.department_name
224 ORDER BY total_admissions DESC;
225
226 -----BASIC COST ANALYSIS (Department-wise)-----
227
228 SELECT
229     d.department_name,
230     SUM(f.treatment_cost) AS total_cost,
231     ROUND(AVG(f.treatment_cost), 2) AS avg_cost_per_patient
232 FROM gold.fact_admission f
233 JOIN gold.dim_department d
234     ON f.department_key = d.department_key
235 GROUP BY d.department_name
236 ORDER BY total_cost DESC;
237
238
239 -----BONUS: MONTH-WISE ADMISSION TREND-----
240
241 SELECT
242     dt.year_number,
243     dt.month_name,
244     COUNT(f.admission_id) AS admissions_count
245 FROM gold.fact_admission f
246 JOIN gold.dim_date dt
247     ON f.date_key = dt.date_key
248 GROUP BY dt.year_number, dt.month_name
249 ORDER BY dt.year_number, admissions_count DESC;

```